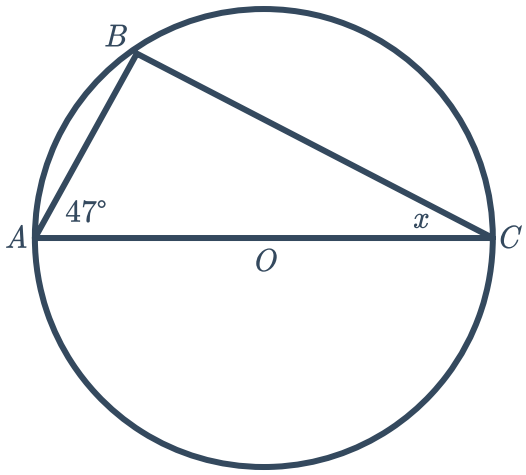
Missing angles

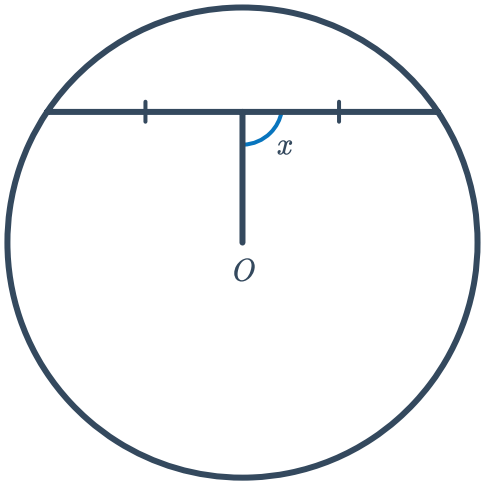
1 In the given diagram, AC is a straight line passing through centre O:

Solve for x.



2 Consider the circle with centre O shown:

Solve for x.

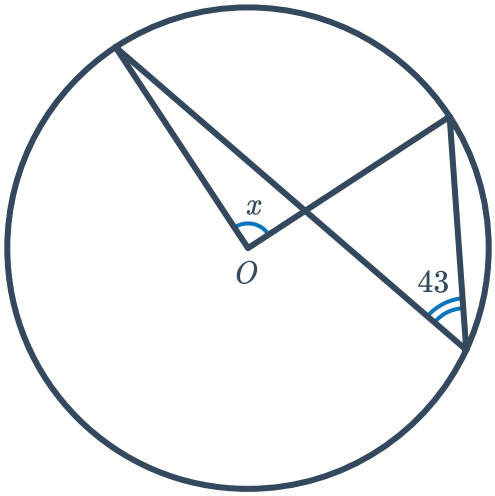


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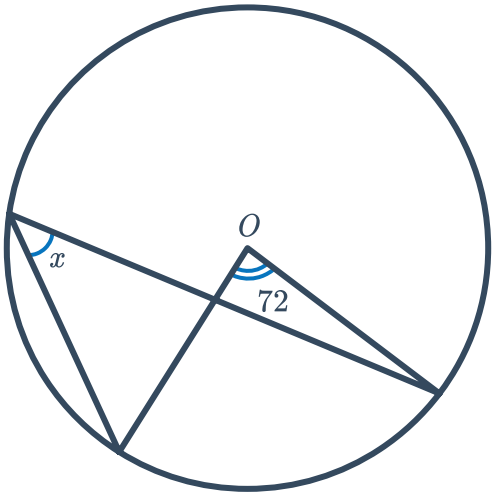
Missing angles (Extended)

3 For each of the following circles with centre O, solve for x:

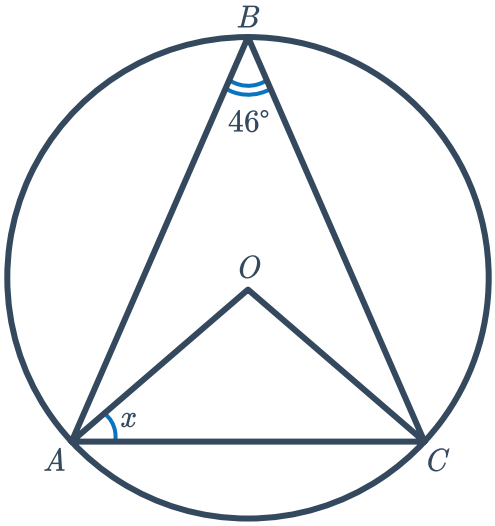
a



b



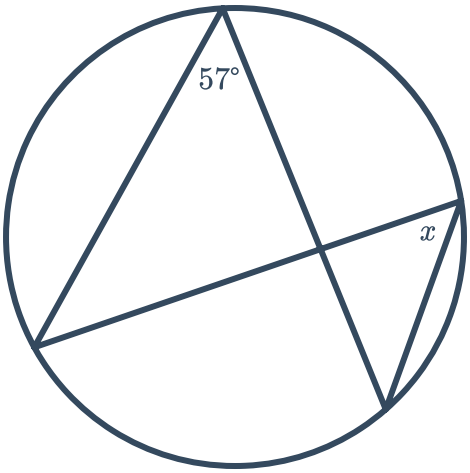
c



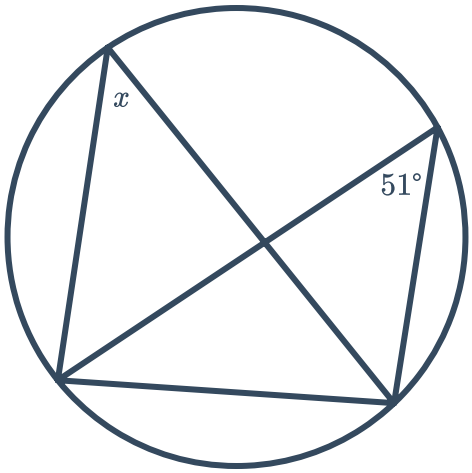
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4 For each of the following circles, solve for x:

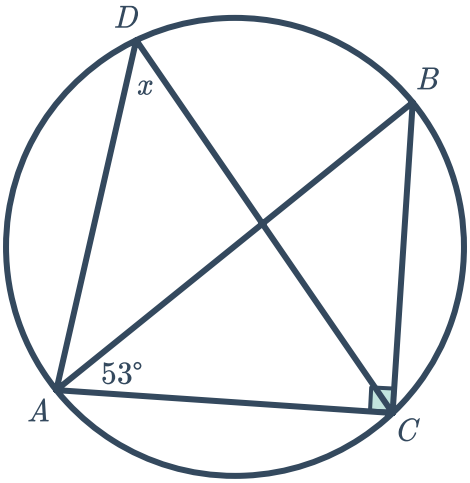
a



b



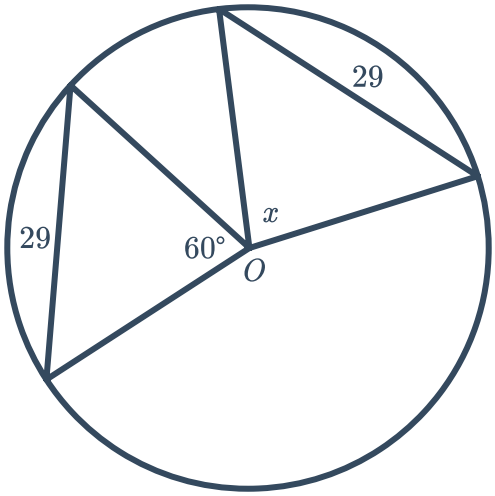
c



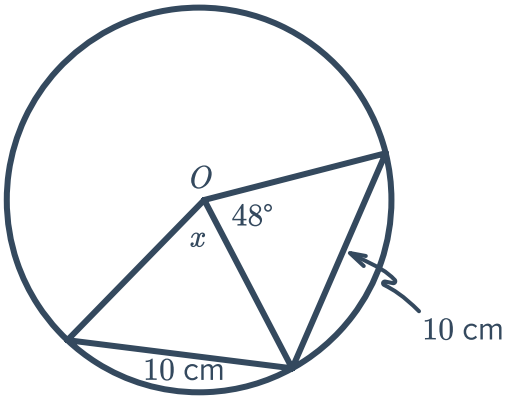
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5 For each of the following circles with centre O, solve for x:

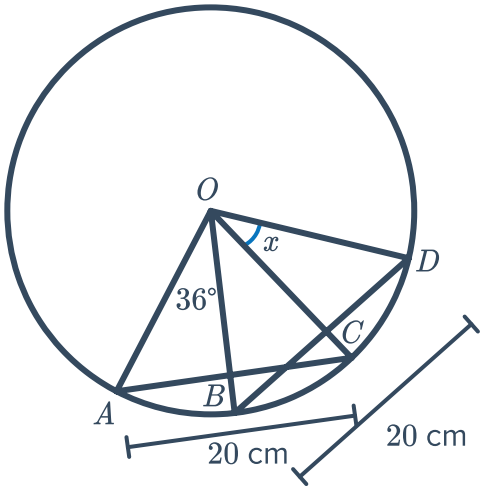
a



b



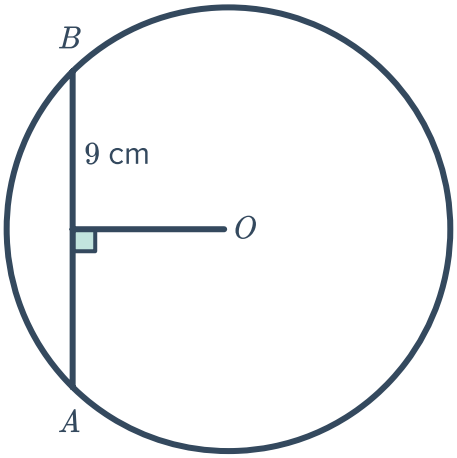
c



Missing lengths

6 Consider the following circle where O is the centre:

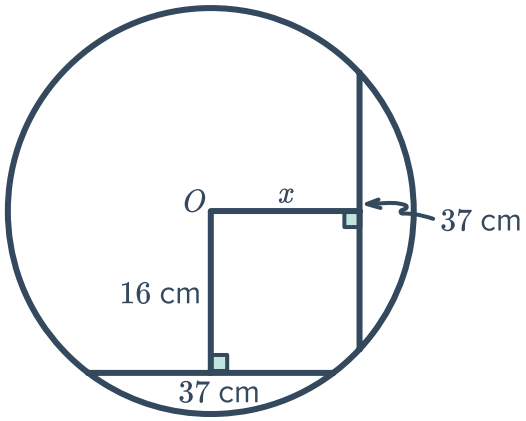
Find the length AB.



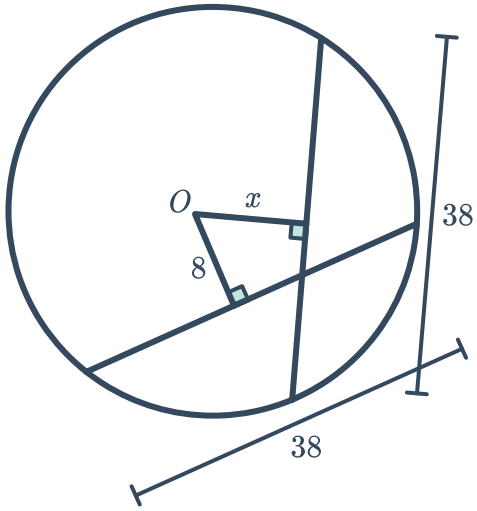
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7 For each of the following circles with centre O, solve for x:

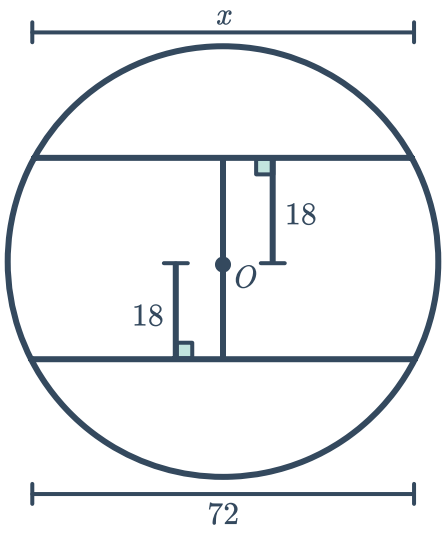
a



b

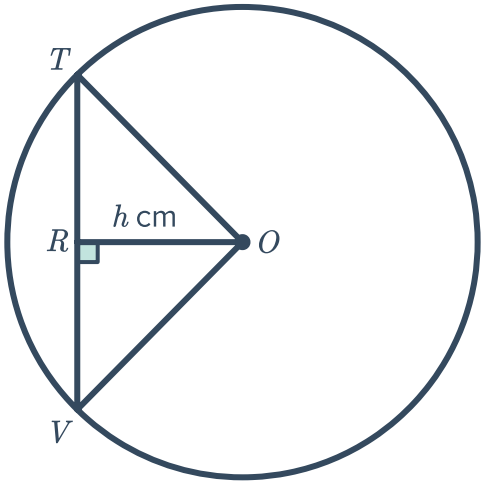


c



8 The given circle has $TR = 12$ cm, the radius is 13 cm, and OR has length h cm.

Find the area of $\triangle TOV$.

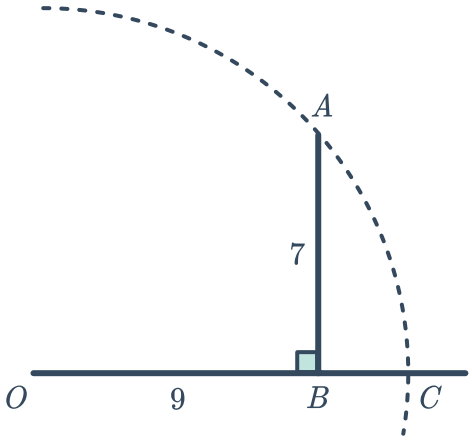


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9 The diagram shows AC as the arc of a circle with O as its centre, $OB = 9$, and $AB = 7$:

a Find the exact length of OA.

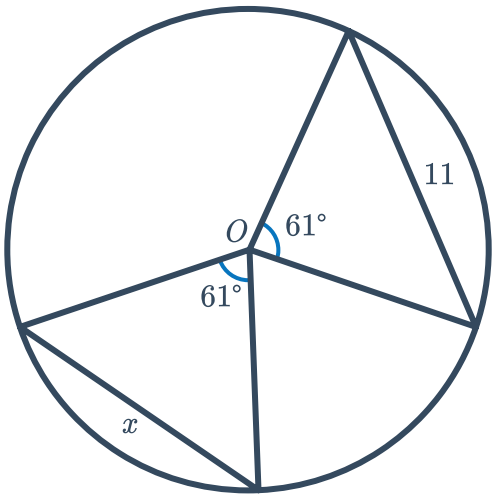
b Hence, find the exact length of BC.



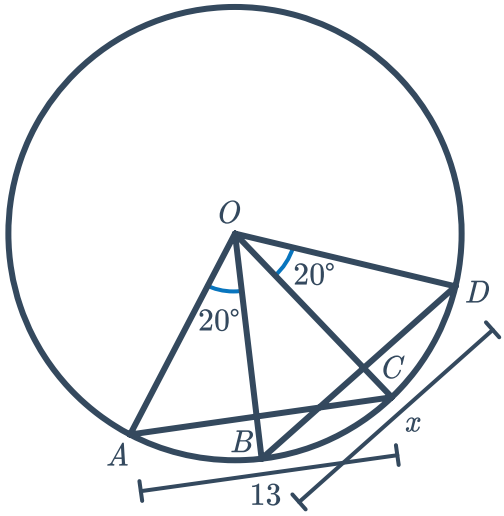
Missing lengths (Extended)

10 For each of the following circles with centre O, solve for x:

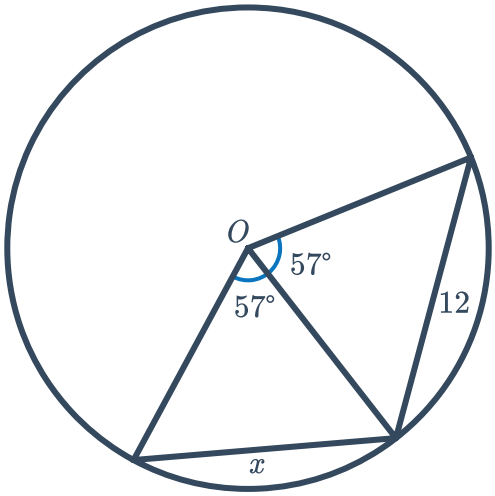
a



b

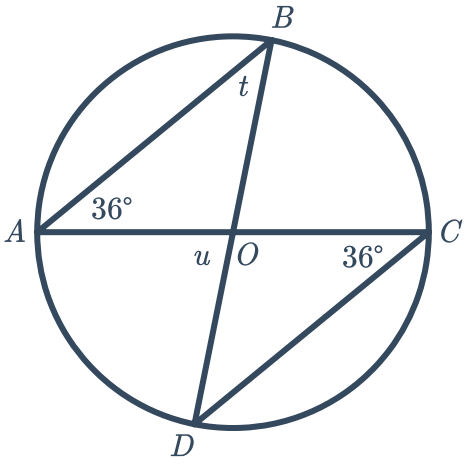


c



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11 Consider the following circle where O is the centre:

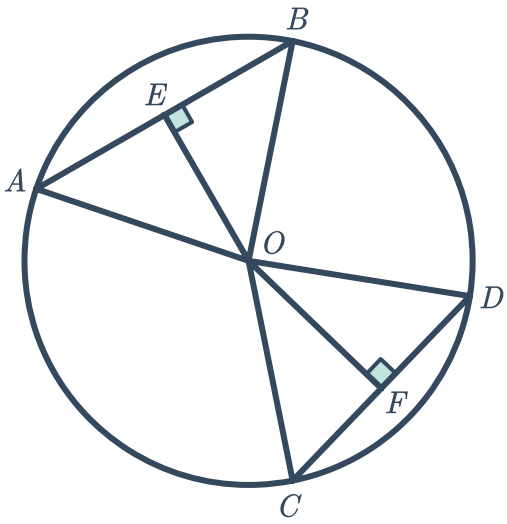


a Solve for t.

b Solve for u.

Proofs

12 Let O be the centre of the given circle with $AB = CD$.



a Prove that $\triangle ABO$ and $\triangle CDO$ are congruent.

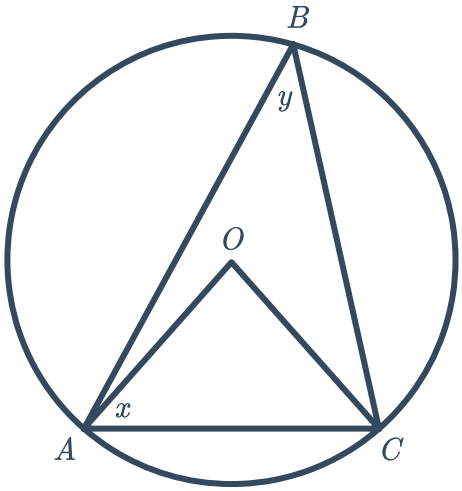
b Hence, prove that $\triangle BEO$ and $\triangle DFO$ are congruent.

13 (Extended)

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Let O be the centre of the given circle:

Prove that x and y are complementary angles.



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